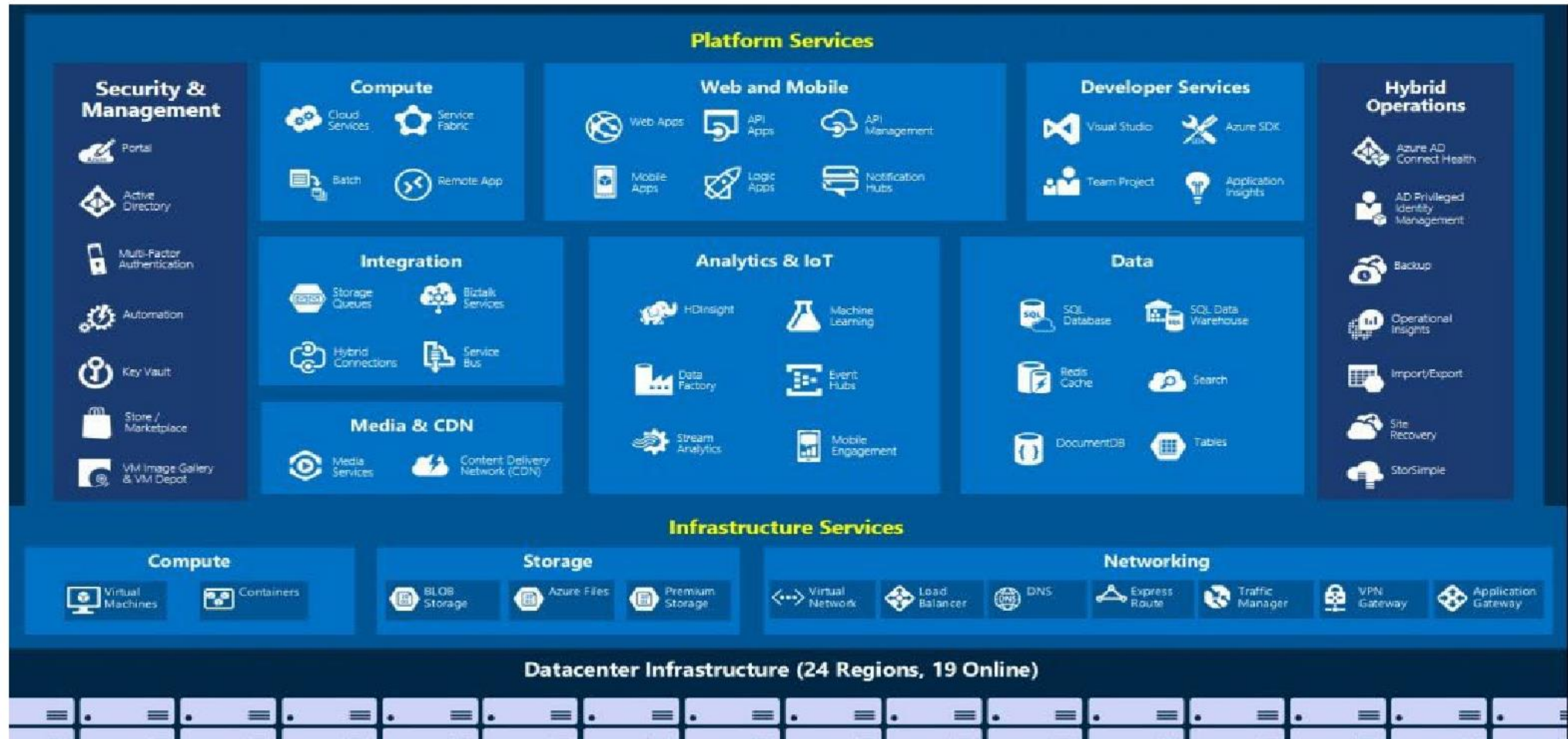


Automatic root cause analysis using LLMs

Feb 10 2026

Cloud services



Productions incidents are costly

The image is a collage of various elements related to production incidents. At the top left, a news headline from Reuters states: "Microsoft investigates Teams outage as services drop for thousands of users", dated July 21, 2022. Below this, a sidebar lists various services: Multi-Factor Authentication, Automation, Key Vault, Store / Marketplace, and VM Image Gallery & VM Depot. The main area features several news snippets. One from Ben Chapman (June 8, 2021) reports that Amazon "missed out on \$34m in sales during internet outage", noting that the e-commerce giant generates \$9,615 in sales per second when its website is down. Another snippet from YouTube (December 14, 2020) states that Google lost \$1.7M in ad revenue during a YouTube outage, affecting services like Gmail. The background is a grid of icons representing different cloud services: Business Applications, End User Computing, Migration & Transfer, Security, Identity & Compliance, Cost Management, Blockchain, Developer Tools, and Management & Governance. A red button labeled "Reliability" is prominently displayed in the lower-left quadrant.

Microsoft investigates Teams outage as services drop for thousands of users
Reuters / Updated: Jul 21, 2022, 10:27 IST

Amazon 'missed out on \$34m in sales during internet outage'
The e-commerce giant generates \$9,615 in sales per second – but not when it's website is down
Ben Chapman • Tuesday 08 June 2021 16:54 • Comments

Google lost \$1.7M in ad revenue during YouTube outage, expert says
YouTube and other Google services, such as Gmail, suffered outage Monday morning

Reliability

Multi-Factor Authentication
Automation
Key Vault
Store / Marketplace
VM Image Gallery & VM Depot

Business Applications
End User Computing
Migration & Transfer
Security, Identity & Compliance
Cost Management
Blockchain
Developer Tools
Management & Governance

Handling incidents

1. Detection and monitoring
2. Diagnosis and root cause analysis
3. Mitigation and post analysis

What are the root causes?

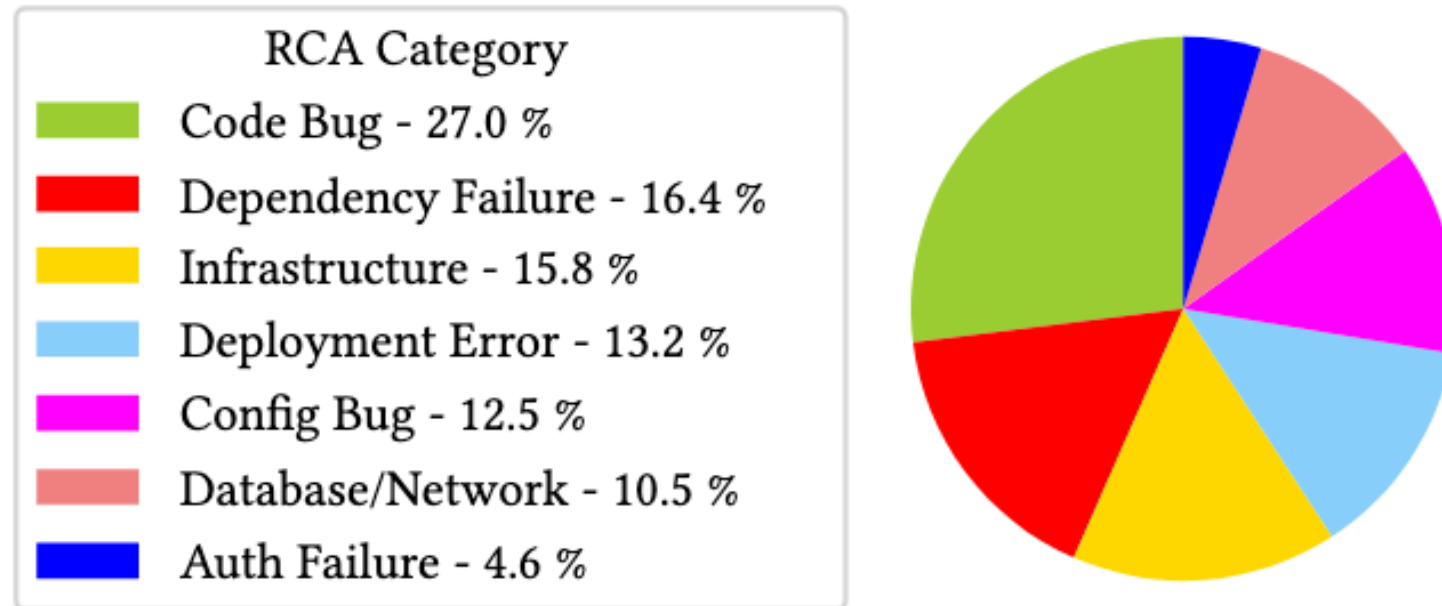


Figure 1: Breakdown of root cause categories

Mitigation steps

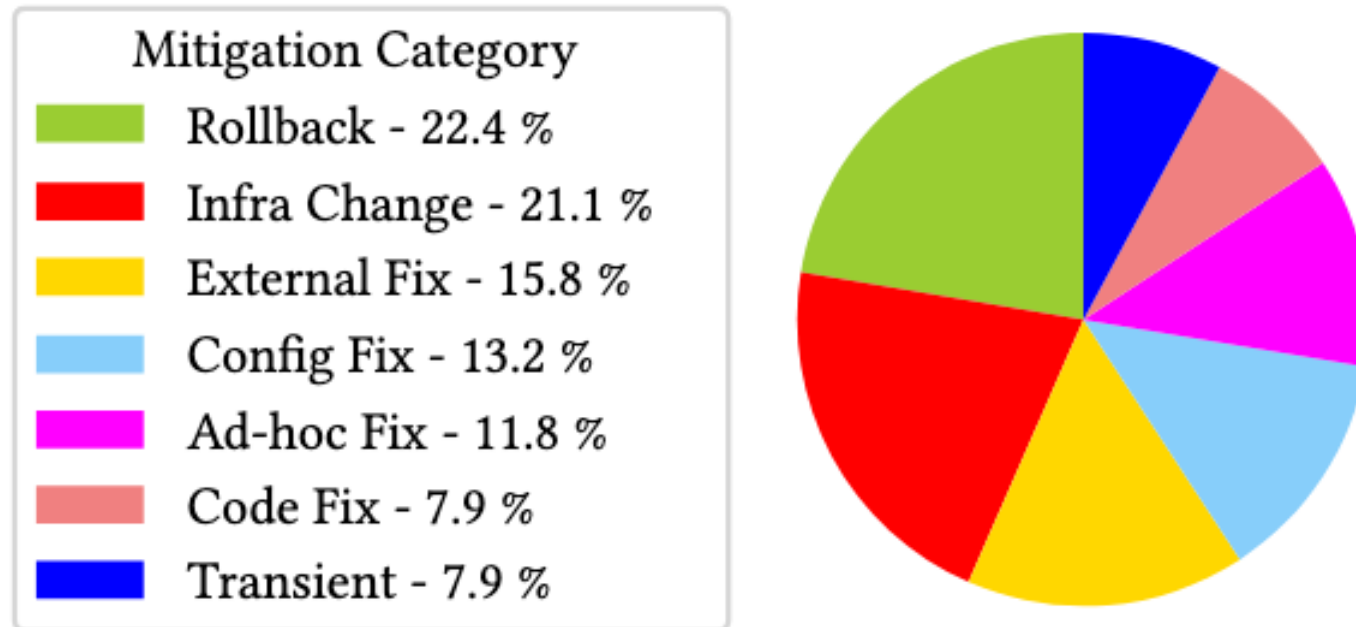


Figure 2: Breakdown of categories for mitigation steps

Detection failures

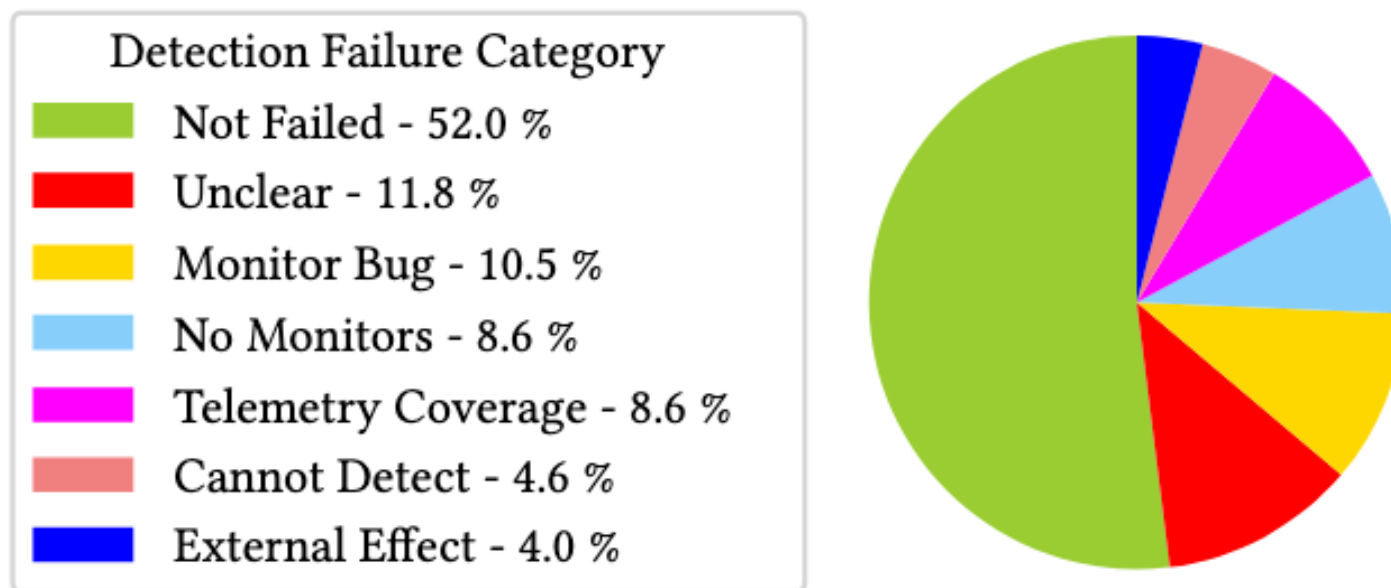
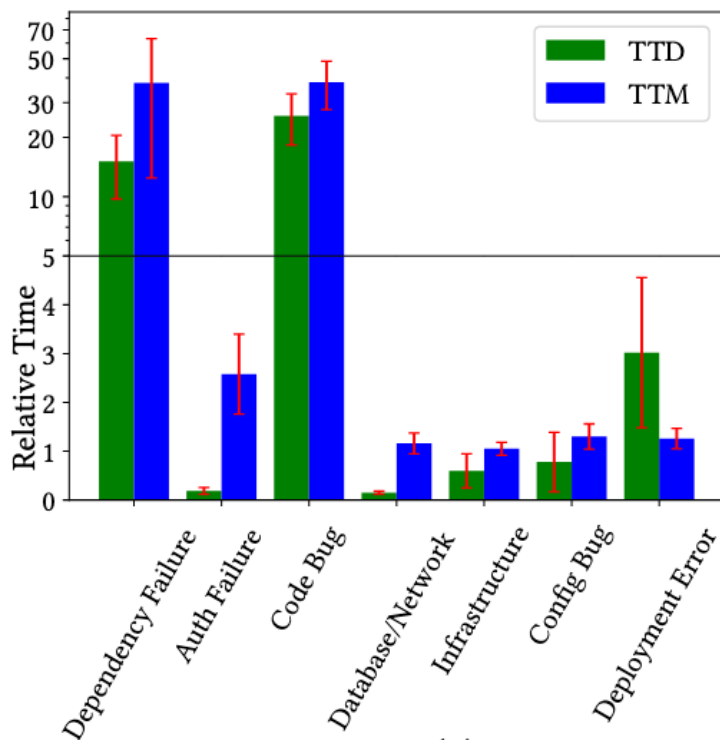
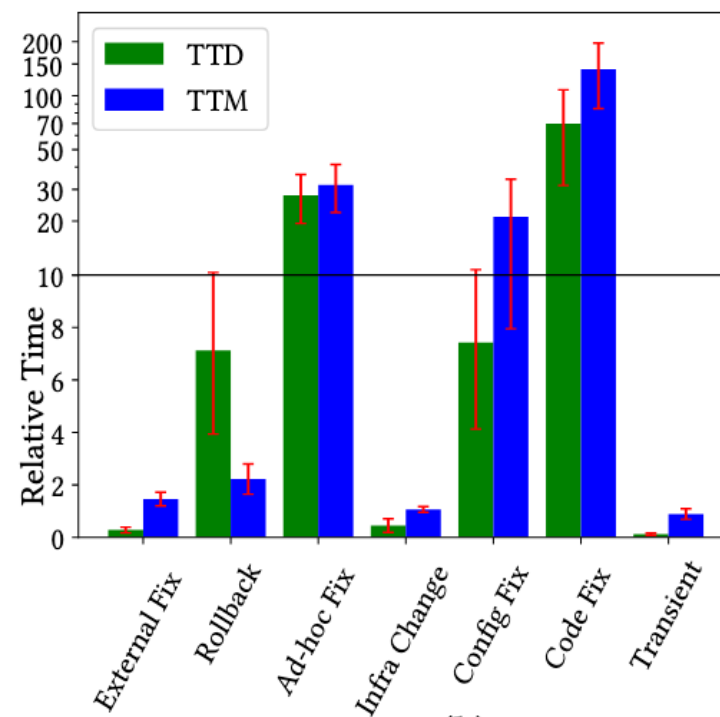


Figure 4: Breakdown of categories for detection failure

Delays in response



(a)



(b)

Figure 3: Average TTD and TTM for (a) different root cause categories; and (b) different categories of mitigation steps. (Y-axis shows the normalized time with the median of time to detect or mitigate of all incidents as 1; for each category, 10% outliers on the high and low side are removed; Y-axis numbers beyond the black horizontal line are shown in log scale).

Other findings

- 21% of incidents where manual effort delayed mitigation, expected improvements in documentation and training
- 25% of incidents where mitigation delay was due to manual deployment steps